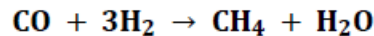


Mechanistic Modeling, Simulation, and Optimization of a Fixed-Bed Ni/Al₂O₃ Reactor for CO Methanation

In this project, you will go beyond the experimental measurement and develop a computational model of the reactor that can predict its behavior under different operating conditions.

The primary reaction is:



Since CO₂ is observed experimentally as an undesired product, the project will also investigate the possible reaction pathway(s) responsible for CO₂ formation and incorporate an appropriate representation into the reactor model.

The model should be based on literature kinetics, thermodynamics, and chemical reactor engineering principles, rather than being purely empirical. Make reasonable assumptions where necessary.

Central Question

Can a computational model be developed from literature kinetics and reactor engineering principles to reproduce the performance of the Ni/Al₂O₃ reactor observed in Experiment #1, and can that model be used to predict and optimize reactor performance beyond the conditions tested experimentally?

Objectives

The project aims to:

- Develop a mechanistic model of the fixed-bed Ni/Al₂O₃ CO methanation reactor.
- Predict reactor performance, including CO conversion and CH₄/CO₂ selectivity.
- Predict species and temperature profiles along the catalyst bed.
- Compare model predictions with the experimental results obtained in Experiment #1.
- Investigate the effects of operating conditions such as temperature, feed composition, flow rate, pressure, and catalyst loading.
- Examine the influence of kinetic parameters and catalyst activity on reactor performance.
- Investigate non-isothermal behavior, temperature rise, and potential hot-spot formation.
- Use the validated model to identify operating conditions that improve reactor performance while maintaining reasonable operating constraints.

Main Components

• Reaction Kinetics and Thermodynamics

Literature investigation of CO methanation kinetics, Ni-based catalysts, equilibrium behavior, and the origin of CO₂ formation.

• Reactor Modeling and Simulation

Development and numerical solution of a one-dimensional fixed-bed reactor model incorporating species and energy balances.

• Experimental Validation

Comparison of computational predictions with the group's experimental GC data from Experiment #1.

- **Sensitivity and Parametric Analysis**

Investigation of how operating conditions and model parameters influence conversion, selectivity, and reactor temperature.

- **Deactivation and Thermal Behavior**

Exploration of catalyst deactivation and the thermal characteristics of the highly exothermic methanation reaction. + (catalyst deactivation - decay model)

- **Optimization**

Use of the validated model to identify favorable operating conditions for achieving high CO conversion and CH₄/CO₂ selectivity while respecting practical constraints.

Overall progression

Experiment → **Modeling** → **Validation** → **Prediction** → **Analysis** → **Optimization**

Expected Deliverables

Each group will submit:

- Well documented computational code.
- A Technical report describing the **model, assumptions, literature basis, validation, results, and limitations.**
- A **Poster** providing a concise overview of the entire project, including the problem/objective, methodology, model, key assumptions, validation, results, and major conclusions.

The final report should include appropriate visualizations such as reactor profiles, model-versus-experiment comparisons, sensitivity analysis, and optimization results.

(Mehrab+Yeamin)	Time to complete the whole work: within 2/3 days
1- Kinetics & Thermodynamics Lead	
- Literature review of CO methanation kinetics on Ni/Al₂O₃ - Compile rate laws + activation energies + pre-exponential factors, with sources - Investigate and propose the CO₂-formation pathway (most likely reverse water-gas shift, $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$, competing with methanation) and its kinetics - Justify why CO₂ is undesired ?	
Deliverable: kinetics/thermo section of report	

(Subah+Somo)	Time to complete the whole work: within 10 september
2 - Reactor Model Builder (Mass & Energy Balances, mainly temp. profile & mat. balance)	
- Set up the 1-D plug-flow (or PBR) mass balance ODEs for each species (CO, H₂, CH₄, CO₂, H₂O) along bed length - Set up the energy balance (non-isothermal, accounting for heat of reaction, possibly heat loss/adiabatic case) - Incorporate pressure drop (Ergun equation) if relevant	
Deliverable: the governing equations, assumptions list (ideal gas, plug flow, no radial gradients, etc.), and the model formulation section of the report	

(Barno+Ashiq)	Time to complete the whole work: within 14 september
3 - Numerical Solution / Code Developer	
- Implement the ODE system numerically (Python/MATLAB) - Handle stiff coupling between mass and energy balances - Produce species concentration and temperature profiles along the bed - Need 2 type of graphs- - Conversion against length of bed - Conversion against weight of catalyst - Build in flexibility to vary T, P, feed ratio, flow rate, catalyst loading as inputs	
Deliverable: well-documented, runnable code (the "computational code" deliverable) + base-case profile plots	

(Shihab+Rifat)	Time to complete the whole work: within 18 september
4 - Validation & Sensitivity/Parametric Analysis	
• Pull the group's actual Experiment #1 GC data (CO conversion, CH₄/CO₂ selectivity) • Compare model predictions vs experimental results; tune/justify any fitted parameters within literature ranges • Run sensitivity analysis: vary temperature, feed composition, flow rate, pressure, catalyst loading - quantify effect on conversion/selectivity/hot-spot behavior Deliverable: validation plots (model vs experiment), sensitivity plots, discussion of discrepancies	

(Ouishi+Rimi)	Time to complete the whole work: within 18 september
5 - Deactivation, Thermal Behavior & Optimization	
• Literature-based treatment of Ni catalyst deactivation (sintering, coking, sulfur poisoning pick what's relevant) and how to represent it in the model • Investigate hot-spot formation / thermal runaway risk given the highly exothermic reaction • Use the validated model to run an optimization (e.g., maximize CO conversion $\times$ CH₄ selectivity subject to a max temperature/hot-spot constraint) Deliverable: deactivation/thermal section + optimization results and recommended operating conditions	

I tried to include everything, along with any extra tasks, in the work distribution. If I've missed anything, please let me know. I'll add it.

The workflow will be like this:

Mehrab and Yeamin will complete their assigned work first. Once they are free, they will help the **next group** complete their work. Then, once those four members are free, all four of them will help the **following groups** with whatever assistance they need.

Note: Some parts of the work are much larger than others, while some are relatively small. At this point, I'm not sure which parts will require the most time or which ones will be easier. Therefore, if any particular part takes longer than expected or seems too difficult, the other members will contribute and help complete it. This way, we can make sure that no one gets stuck with an excessive workload and that everything is completed on time.

We have three weeks to complete everything. However, it would be quite a hassle to check everyone's work and merge everything at the last moment. So, our plan is to complete the entire work by **19 September (please do keep this in mind, also we're already one week behind)**. This will give us one full week to thoroughly check everything, correct any mistakes and most importantly, let **Farhatul Abrar Sir** review our work, as he will be giving us a major portion of the marks.

Another important thing is that we also have to prepare a **poster presentation**. Therefore, we need to **keep the entire final week for designing and preparing the poster properly.**

